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Data-Driven Platform Design: Patent Data and Function Network Analysis

A properly designed product-system platform seeks to reduce the cost and lead time for design and development of the product-system family. A key goal is to achieve a tradeoff between economy of scope from product variety and economy of scale from platform sharing. Traditionally, product platform planning uses heuristic and manual approaches and relies almost solely on expertise and intuition. In this paper, we propose a data-driven method to draw the boundary of a platform-system, complementing the other platform design approaches and assisting designers in the architecting process. The method generates a network of functions through relationships of their co-occurrences in prior designs of a product or systems domain and uses a network analysis algorithm to identify an optimal core–periphery structure. Functions identified in the network core co-occur cohesively and frequently with one another in prior designs, and thus, are suggested for inclusion in the potential platform to be shared across a variety of product-systems with peripheral functions. We apply the method to identify the platform functions for the application domain of spherical rolling robots (SRRs), based on patent data.
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1 Introduction

Mass customization has become an important competitive edge of companies. To satisfy diverse market niches, companies develop product families based on product platforms to increase product variety with reduced cost and lead time [1,2], in addition to other driving factors such as product flexibility [2–5]. Typically, a product family comprises a group of products that are derived from a product platform by either adding, reducing, or substituting some features and functionality or scaling design parameters in the product platform [2–5]. A product platform is defined as a collection of common elements, components, subsystems, and interfaces repeatedly used in a variety of products or systems [6–9].

A well-designed product platform is the heart of a successful product family. However, as pointed out in Ref. [10], planning or defining the product platform is one of the most challenging tasks in product family design for companies. In the previous studies, designers define a product platform mainly by identifying modules or scalable design parameters from product architectures known a priori. Since the architectures are typically constructed based on heuristics, expertise, and intuition, there exists uncertainty in how to identify which functions should be included ex ante in a platform, where the corresponding analytic approaches are underdeveloped [11]. Moreover, the existing approaches may give excessive attention to physical structures and design parameters, but overlook customer needs represented by product features and functions during platform planning.

In this paper, we develop a data-driven method to draw the boundary of the potential platform by analyzing the functions and their co-occurrence relationships in prior designs from a product or systems domain. Particularly, the method leverages the core–periphery structure theory and detection algorithm from the complex network analysis (or graph theory) literature to detect the optimal core–periphery partition of the function co-occurrence network. The resultant network core suggests a group of functions that constitute a potential product platform, and individual or subset of the peripheral functions can be combined to generated family products.

Therefore, our research aims to contribute to data-driven design research by a network analysis method for function-based platform design. In the following, we review the related work, introduce the method, and then apply it to identify potential function platforms for the application domain of spherical rolling robots (SRRs) based on patent data.

2 Related Work

2.1 Platform Planning. Product and systems platform planning is a key of product family design. The success of platform planning lies in the artful balance between commonality and distinctiveness [10]. In the literature, there are two major approaches to define a product platform, namely the module-based (or configurational) approach [4,12] and the scale-based approach [2]. The module-based approach starts with identifying modules in product architectures [4,13] using various methods and techniques [14–19]. Based on the identified modules, designers construct a basic modular platform shared across a product family; from the platform, they add, remove, or substitute modules to generate product variants with functional diversity. For the scale-based approach, defining a product platform is to select design

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parameters that take common values, which is followed by or goes in parallel with determining the optimal values for the common and scaling design parameters to achieve performance diversity [20]. Typically, the tradeoff between commonality and performance and the sensitivity of performance to scaling variables are analyzed for product platform identification [2,21–23].

Most platform identification approaches reported in prior product family design studies define platforms based on given product architectures and assume the functions or components underlying the architectures are known a priori [21,24–28], according to expertise and intuition. In order to draw the boundary of a platform, several analytic methods have been proposed to select the functions, components, or design parameters to be made common and included in the platform. In particular, platform identification may be formulated as a combinatorial optimization problem with constraints that search the optimal commonality with acceptable product performance from all combinatory possibilities [24,29–34]. Among them, some methods have applied combinatorial algorithms, such as genetic algorithms, to solve the large-sized combinatorial problems [29,32–36]; others simultaneously determined modular design and platform planning using multi-objective optimization [30,31]. Researchers also defined the component shareability [23] and design variable sensitivity [37] by which the components or design variables can be sorted to compromise the commonality optimization problems for reducing the problem sizes.

Some other methods attempt to identify the common functions or components by analyzing user requirements. Kurtadikar and Stone [11] observed that customer needs with higher weight rated by customers and lower frequency in customer need statements constitute the core needs for architecting a product platform. Agard and Kusiak [38] classified customers based on their similarity in behaviors, and derived association among the requirements of a selected group of customers as a basis for functional and technical design of a platform. Kuang and Jiang [39] identified design parameters that have little influence on customers' affective evaluation on product representations as platform parameters based on Kansei Engineering [40]. Cheng [41] classified customer needs in to basic, expectable, and adjunctive functional requirements and defined design parameters achieving the basic function requirements as common platform parameters.

Another strand of methods extracts common structures and components with certain properties to conceive product platforms from a group of existing variants. A group of researchers [42,43] utilized clustering analysis based on similarities between elements or design variables to identify common elements or design variables for architecting a product platform. Design structure matrix analysis is employed for this purpose. Kalligeros et al. [44] employed a sensitivity DSM (S-DSM) to identify design variables with low sensitivity as platform variables. Steva et al. [45] integrated DSM with bill-of-material, and function diagrams, function-component matrices, and product-vector matrices, respectively, to identify common platform elements. In addition, Qu et al. [46] adopted a graph-theoretic approach to represent product variants and used genetic algorithms to identify the largest structure commonality as the product platform. Kim and Moon [47] defined a few component property indicators and suggested components with high sustainability and commonality and low risk for sustainable product platform design.

Generally speaking, the prior platform identification methods typically rely on the knowledge of small samples of existing product variants or user requirements to identify the common functions, components, or design parameters. Data-driven platform identification, especially based on function analysis, are still underdeveloped. This paper aims to address this gap by presenting a network analysis method for product platform identification based on function information in massive prior design data. In the network of functions with interfunction links weighted according to their co-occurrences in prior designs, the platform functions would appear in the cohesively connected core of the network as they are shared across many product variants. Therefore, we detect the core–periphery structure in the function co-occurrence network to reveal the core functions to inform platform identification.

2.2 Core–Periphery Structure and Methods. In a network exhibiting a core–periphery structure, the core refers to a set of nodes that are at the center and cohesively connected to each other, while the periphery refers to a set of nodes that are around the center and loosely connected to the core [48–50]. Figure 1(*a*) shows an ideal core–periphery structure, i.e., nodes in the core are fully connected, whereas nodes in the periphery are only connected to the core but not connected to each other. Since their introduction in the 1950s, several types of core–periphery network structures have been observed and studied, such as rich-clubs [51,52], nested network structure [53–55], bow-tie network structure [56–58], and onion network structure [59–61]. Examples for these different types of core–periphery networks are also shown in Figs. 1(*b*)–1(*e*).

The core and the periphery play different roles in the functioning of a complex system. For example, in social networks, players in the core take more important social roles and are more influential than those in the periphery in a community [50,62]. In cellular and flow-type networks, the core is the more degenerate and robust part with lower variability and evolvability [63]. Core–periphery network structure has also been studied in the engineering field to analyze the stability and robustness of complex systems, such as air transportation [64] and power grids [65]. The core–periphery network structure also provides a conceptual lens to identifying the functions for the product platform and the functions for variants in the product family.

There are two main types of models for detecting core–periphery structures in networks: discrete and continuous models. The discrete models partition the network nodes into a core class and a periphery class by maximizing a core–periphery coefficient that characterizes the structural distinction between the two classes. Borgatti and Everett [50] operationalized the core–periphery coefficient as the correlation between the adjacency matrices of an empirical network and an ideal core–periphery network as a benchmark. Alternative benchmark networks have been used to detect different core–periphery structures in directed and undirected, reflexive and nonreflexive networks, and so forth. Besides maximizing the correlation, alternative approaches were proposed to minimize the Hamming distance, or maximize density in the core and minimize density in the periphery [50]. Researchers have employed simulated annealing [66,67], Tabu search [68], differential evolution [67,69], and genetic algorithms [70] to solve such combinatorial optimization problems. In addition, other definitions of the core–periphery coefficient have also been proposed, and interested readers may refer to Holme [71] and Csermely et al. [48].

For networks that by nature do not have a clear core–periphery structure, the continuous models are more suitable, which assign a coreness score to each node rather than a dichotomous classification [50]. The coreness scores are determined by finding a vector such that the product of the vector and its transpose (i.e., the pattern matrix) is as close as possible to the empirical adjacency matrix. Elements constituting the vector are coreness scores of corresponding nodes. According to the definitions of closeness between two matrices, the continuous models may choose to optimize the correlation coefficient, squared difference, or residual between them, respectively [50,72]. The standard Fletcher-Powell [73] function maximization procedure and simplex algorithms [74] can be used for these optimization problems. In addition, researchers also proposed other definitions and calculating methods of coreness, and interested readers may refer to Rombach et al. [75] and Csermely et al. [48].

The discrete and continuous models have merits and limitations, respectively. The discrete models offer straightforward

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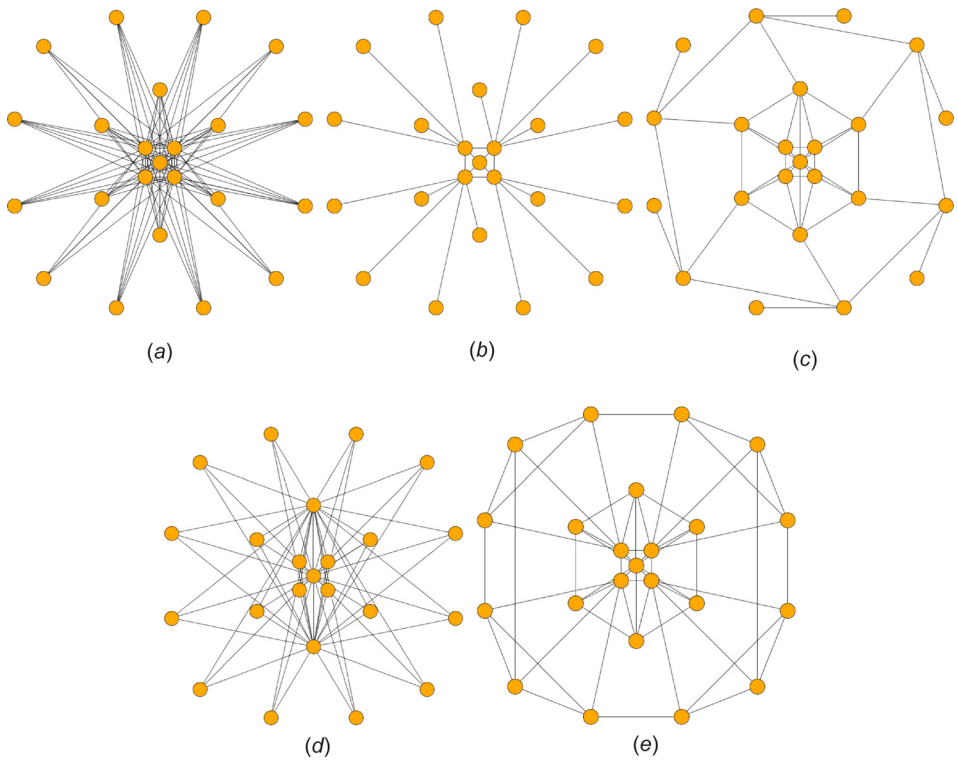

Fig. 1 Structures of various types of core–periphery networks: (*a*) Typical core-periphery, (*b*) Rich-clubs, (*c*) Nested, (*d*) Bow-tie, and (*e*) Onion

stopping criteria to determinate a specific partition. However, the dichotomous partition might be over-simplified and arbitrary, when the network by nature does not embed a core–periphery structure or a sharp boundary between the core and the periphery. The continuous models give a coreness score to each node, which provides more reference to weigh the nodes and adjust the core–periphery division based on other considerations. But the partition is heuristic and not deterministic.

The core–periphery network structure theory and related detection algorithms were first developed in the field of complex networks and have not been previously adopted for engineering design and research of the type reported in this paper. Meanwhile, this paper fills the gap by applying the core–periphery structure theory and detection algorithms to the empirical function co-occurrence network to identify core and peripheral functions, and thus, inform the design of the product platform and product variants in a family. This linkage is possible because of the platform-variants relationships in a product family resonant with the core–periphery structure in the function co-occurrence network.

In turn, our research on data-driven platform design adds to the growing studies on data-driven design, which normally utilize data science methods such as decision tree [76], network analysis [77], neural networks [78], clustering [76,79,80], and regression [81] to discover design knowledge [77,82,83], gauge user needs and market trends [76,80,83–85], and support design ideation [83,86] and product configuration [85,87–89]. In the following, we introduce the details of our method.

3 Our Method

In this section, we introduce a data-driven method to identify core functions for platform design. The method is based on a core–periphery analysis on the function co-occurrence network, which is constructed using empirical data of functions extracted from prior designs in a product domain. The originality of the method resides in using a function co-occurrence network to represent the design knowledge extracted from prior designs and detecting the core–periphery structure in networks to identify the core and peripheral functions to inform the function design of product platforms and variants. The method includes two primary steps.

3.1 Step 1: Construct Function Co-Occurrence Network Based on Prior Designs in a Product Domain. Functions in a product platform are shared among variants, so they have high co-occurrence with each other and other differentiating functions. In this step, data on prior designs in a product domain of interest are required to capture the co-occurrence relationships between functions. First, we extract functions included in each prior design to build the full function set that contains all functions occurring in prior designs; then, we count the co-occurrence in the prior designs of each pair of these functions to generate their co-occurrence matrix, based on which the function co-occurrence network is constructed. In such a network, each node is a function that appeared in any prior design; the link weight indicates that the frequency a pair of functions has co-occurred in prior designs. Herein, we focus on the primary functions[2] that explain why a product exists and embody the value of a product to customers, instead of the secondary functions that realize primary functions at the lower level. In our later case study, we utilize patent data in a product domain, whereas one can also retrieve and analyze functions from other types or sources of data on prior designs.

3.2 Step 2: Detect Optimal Core–Periphery Network Partition to Identify Core Functions for Inclusion in a Platform. Platform functions are shared by many product variants, and thus, can be captured in the core of the function co-occurrence network via a core–periphery structure analysis. Because the function co-occurrence network is undirected and

[2]Primary functions can be identified by customers, users, or stakeholders at the product or system level, whereas secondary functions are defined at a more specific level, which leads to more specific technologies or principles. A primary function can be decomposed into multiple secondary functions [90].

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nonreflexive, we use the most common core–periphery benchmark network (i.e., 1 s for core–core links but 0 s for periphery–periphery links, while ignoring core–periphery links), and the most common model (i.e., the correlation model), to find an optimal core–periphery partition. The optimization function is given by Eq. (1) together with constraints in Eq. (2) for the discrete model and Eq. (3) for the continuous model [50]. Herein, Eq. (1) maximizes the normalized Pearson correlation coefficient between the adjacent matrices of the empirical function co-occurrence network and the benchmark network

$$\max \rho = \frac{\sum_{1\le i\le n}\sum_{i<j\le n}(a_{ij}-\overline{a_{ij}})\cdot\left(\delta_{ij}-\overline{\delta_{ij}}\right)}{\sqrt{\sum_{1\le i\le n}\sum_{i<j\le n}(a_{ij}-\overline{a_{ij}})^2\cdot\sum_{1\le i\le n}\sum_{i<j\le n}\left(\delta_{ij}-\overline{\delta_{ij}}\right)^2}} \tag{1}$$

s.t.

$$\delta_{ij} = \left\{\begin{array}{l} 1 \text{ if node } i \text{ and node } j \text{ are in the core} \\ 0 \text{ if node } i \text{ andnode } j \text{ are in the periphery} \end{array}\right\} \tag{2}$$

or

$$\delta_{ij} = c_i c_j \tag{3}$$

where a_{ij} denotes the link weight between functions i and j (i.e., the co-occurrence frequency between them) in the function co-occurrence network and $\overline{a_{ij}}$ is the mean, δ_{ij} and $\overline{\delta_{ij}}$ denote the corresponding counterpart in the benchmark network and its mean, and c_i is the coreness score of function i. If we constrain c_i to 0 s and 1 s, Eqs. (2) and (3) are the same.

To solve the optimization problem, we employ the widely adopted network analysis software UCINET [91] to detect the optimal core–periphery partition of the function co-occurrence network and calculate the continuous coreness scores of individual functions. UCINET uses a steepest ascent hill climbing algorithm[3] [92] and a Nelder-Mead simplex method[4] [74], respectively, for the discrete and continuous models. Although these heuristic algorithms present high computational efficiency, they may return local optimum to the core–periphery partitions.

As a significance test, we calculate the degree to which the function co-occurrence network shows a core–periphery structure compared to random networks generated by shuffling links in the function co-occurrence network. We use the z-score of the obtained core–periphery coefficient for this purpose, which is calculated as

$$z = \frac{\rho - \mu}{\sigma} \tag{4}$$

Herein, ρ is the core–periphery coefficient obtained for Eq. (1), μ is the mean core–periphery coefficient of an ensemble of networks having the same link weight sequence as the function co-occurrence network, and σ is the corresponding standard deviation. We approximate μ and σ by the mean and standard deviation of a set of random networks generated by randomly shuffling links in the function co-occurrence network. If the z-score is positive and larger than 2, the empirical network has a significantly higher core–periphery coefficient than the null model at 5% level.

[3]The optimization begins with a random assignment of nodes to the core or the periphery and then evaluates the correlation between the empirical adjacent matrix with the ideal adjacent matrix. Then, for each node, it sees if the correlation can be improved by switching the node to the other class. This is repeated until no more improvements are possible.

[4]The optimization begins with a random assignment of coreness scores between 0 and 1 to each node to construct the initial working simplex S and then evaluates the correlation between the empirical adjacent matrix and the ideal adjacent matrix. Then, it transforms the working simplex to see if the correlation can be improved. This is repeated until no more improvements are possible.

Fig. 2 (*a*)–(*c*) The spherical rolling robot product family from Sphero, Inc., and (*d*) Virgo from SUTD: (*a*) Sphero, (*b*) BB-8, (*c*) BB-9E, and (*d*) Virgo

Then, according to the obtained core–periphery partition, designers are recommended to consider functions in the core for a product platform design and functions in the periphery for generating variants in a product family. The resonance between the platform-variants relationships in a product family and the core–periphery structure in the function co-occurrence network enables the core–periphery analysis to identify core functions for product platform design. To the best of our knowledge, this paper is the first to leverage the core–periphery network structure and corresponding algorithms to analyze the function co-occurrence network for product platform and variant family design.

4 Case Study of Spherical Rolling Robots

In this section, we demonstrate the proposed method via a case study of SRRs and discuss the product family of Sphero, Inc., as an example of SRR platform design. SRRs are spherical shape robots that can propel themselves to roll around on the ground. Typically, the movement of SRRs is driven by a cart [93,94] or a pendulum system [95,96] contained in the shell or other mechanisms [97–100]. SRRs have a great potential for utility in defense and consumer markets, with increased functionality and capability [101–105]. Figure 2 presents four SRR designs.[5,6] The first three are the product family developed by Sphero, Inc. (Boulder, CO), and the last one is Virgo developed at Singapore University of Technology and Design (SUTD, Singapore) with extreme intelligence, surveillance and reconnaissance and autonomous capabilities [102].

4.1 Step 1: Construct Function Co-Occurrence Network. We analyzed a patent set of 153 SRR patents from an exhaustive SRR patent search from 1790 to May 31, 2016 from a prior publication [77] to extract function data. We manually identified the primary functions that occur in each SRR design disclosed in these patents. Then, we manually categorized the function terms describing the same or similar functions and named them using a more common term. As a result, we identified 87 unique function

[5]https://www.sphero.com/
[6]https://www.sutd.edu.sg

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Table 1 The full list of functions identified from SRR patents

Function	Occurrence	Coreness (rank)	Function	Occurrence	Coreness (rank)
Roll on ground	150	0.391(1)	Actuate accessory	8	0.027(42)
Convert energy	150	0.391(1)	Appoint movement	7	0.02(46)
Store energy	150	0.39(3)	Fly	7	0.015(49)
Control direction	91	0.293(4)	Transform	7	0.02(46)
Remote control	67	0.239(5)	Harvest natural energy	6	0.018(48)
Process data	45	0.186(6)	Increase movability in water	6	0.012(50)
Store data	42	0.182(7)	Move eccentrically	6	0.008(52)
Self-balance	39	0.14(11)	Switch on/off automatically	5	0.008(52)
Transmit data	38	0.17(8)	Stabilize	4	0.01(51)
Plan path	37	0.163(9)	Traverse chasm	4	0.008(52)
Photograph surrounding	35	0.153(10)	Increase adhesive force	4	0.007(56)
Propel in water	31	0.083(28)	Orient camera	3	0.007(56)
Detect acceleration	29	0.135(12)	Thermal control	3	0.008(52)
Measure environment	28	0.135(12)	Resist blast	2	0.003(65)
Track position	27	0.13(14)	Clear mine	2	0.003(65)
Produce sound	27	0.111(16)	Dispose payload	2	0.005(60)
Detect angular velocity	26	0.127(15)	Dock itself	2	0.007(56)
Resist water	26	0.074(34)	Mark environment	2	0.003(65)
Detect inner part orientation	22	0.109(17)	Passenger control	2	0.002(69)
Emit light	21	0.097(19)	*Follow path*	*2*	*0.007(56)*
Communicate information	19	0.103(18)	Track target	2	0.005(60)
Locate target	*18*	*0.094(20)*	Fold itself	2	0.004(62)
Display image/data	18	0.091(23)	Sense weight	2	0.002(69)
Map surrounding space	18	0.094(20)	Board easily	2	0.004(62)
Avoid obstacle	*17*	*0.044(35)*	Aim actuator	1	0.001(76)
Play music	17	0.085(25)	Give out smell	1	0(85)
Sense orientation	17	0.093(22)	Recode angular position	1	0.002(69)
Transport passenger	17	0.037(38)	Obtain annular view	1	0.001(76)
Detect energy level	15	0.086(24)	Park on slope	1	0.002(69)
Record time	15	0.085(25)	Climb	1	0.001(76)
Wireless charge	*15*	*0.085(25)*	Detect metal	1	0(85)
Shake/vibrate	14	0.077(29)	Fire	1	0.001(76)
Detect collision	13	0.077(29)	Sense ground contact	1	0.001(76)
Control by gesture/voice	13	0.075(31)	Integrate optical signal	1	0.001(76)
Dampen shock	13	0.041(36)	Pick up litter	1	0.001(76)
Modulate light	13	0.075(31)	Pivot external accessory	1	0.004(62)
Detect status	13	0.075(31)	Process image	1	0.002(69)
Traverse obstacle	13	0.04(37)	Recover braking energy	1	0.001(76)
Hold external accessory	12	0.03(40)	Dispose chemical pollutant	1	0.003(65)
Carry payload	11	0.027(42)	Guarantee safety	1	0(85)
Jump/hop/bounce	10	0.025(44)	Sense pressure	1	0.002(69)
Brake	9	0.021(45)	Detect velocity	1	0.001(76)
Detect inclination	9	0.03(40)	Assist in therapy	1	0.002(69)
Detect obstacle	*9*	*0.035(39)*			

categories from the prior SRR designs disclosed in the patent set. Each function category was named following the verb-object or verb format [90,106]. Table 1 lists the full set of function categories, as well as their occurrences in the patents (i.e., the count of SRR patents in which a function is identified). Such occurrences indicate the popularity of a function in the prior SRR designs. Herein, only the primary functions are identified and analyzed. For example, the function "Roll on Ground" can be decomposed into a few secondary or more generic functions, such as transmit energy, change speed, and transmit motion, which are not identified and analyzed.

Figure 3 visualizes the function co-occurrence network. Each node represents one of the 87 functions listed in Table 1. The node size corresponds to the occurrence of the function. The link weight indicates how frequently the corresponding pair of functions co-occurs in prior SRR designs, i.e., the number of patents in which a pair of functions co-occur. In the network, functions frequently co-occurring with others, i.e., nodes having more strong links, form a cohesively connected core. This group of functions in the network core is more likely to jointly appear with varied combinations of functions in the periphery in a high number of prior SRR

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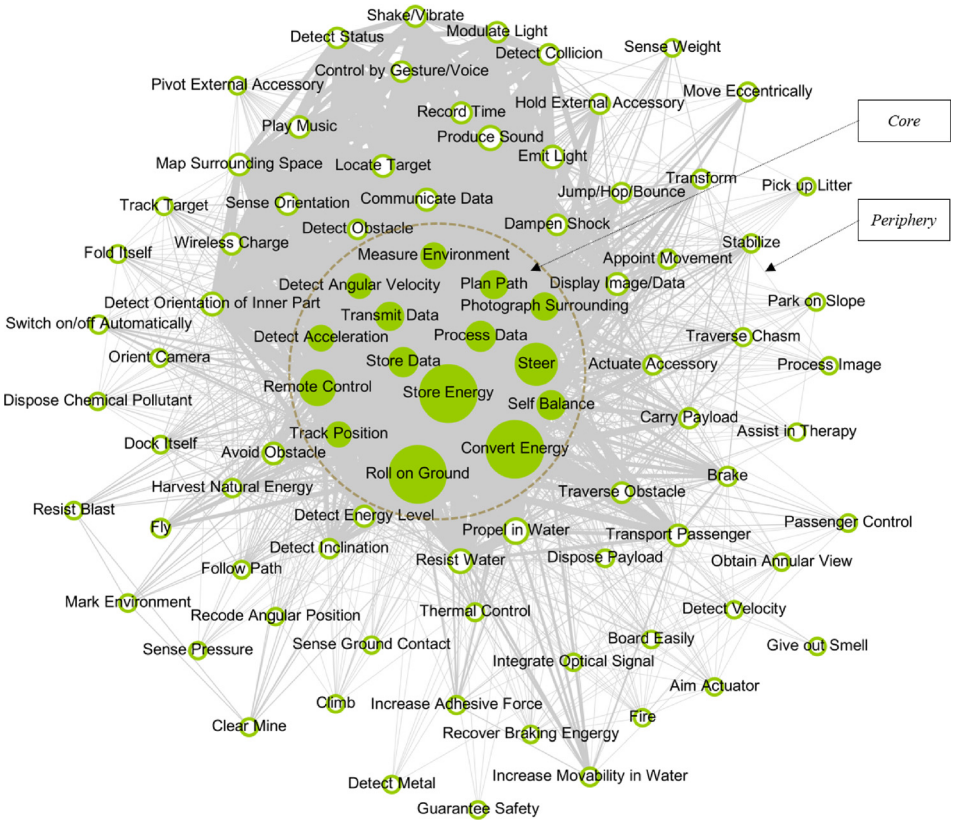

Fig. 3 The co-occurrence network of functions identified in prior SRR designs. Core functions are highlighted as solids and peripheral functions are represented as circles.

designs and should be considered for inclusion in potential SRR platforms.

4.2 Step 2: Detect Optimal Core–Periphery Partition. The method described in Sec. 3 was applied to find the optimal core–periphery partition of the function co-occurrence network. The resultant core includes 15 functions, which are underlined in Table 1 and highlighted as solid nodes in Fig. 3. The rest are peripheral functions represented as circle nodes in the outskirt of the network in Fig. 3. The corresponding core–periphery coefficient to the optimal partition is 0.528,[7] indicating a strong but far from perfect fit with the ideal benchmark. A robustness test using Hamming distance between the matrices of the empirical and the benchmark networks gives a similar partition.

Moreover, this empirical core–periphery coefficient value is significantly higher than the average core–periphery coefficient (0.143) of an ensemble of 100 comparable random networks generated by shuffling links in the function co-occurrence network. The z-score is 16.1934, which indicates that the relatively high value of core–periphery coefficient is significant at 5% level. That is, the empirical function co-occurrence network presents a core–periphery structure to a significant degree. And the discrete model well captures its structure.

More detailed descriptions of the 15 core functions are provided in Table 2, together with choices of components to fulfill these functions. Among the 15 core functions, Roll on Ground, Convert Energy, Store Energy, Control Direction, and Self-Balance are required to propel a robot, while the others are generally related to controlling a robot and monitoring the environment. These functions constitute the basic functionality of a general autonomous and sensing mobile robot, and thus, should be included in a platform. Then, a family of SRR products for specific function purposes can be developed via integrating some peripheral functions with the platform.

For example, designers may design an autonomous SRR by integrating some peripheral functions, such as Detect Obstacles, Avoid Obstacles, Locate Target, Map Surrounding Space, and Follow Path, with the platform functions. To develop recreational SRRs, designers can consider other peripheral functions, such as Play Music, Emit Light, Modulate Light, and Shake/Vibrate, on top of the platform functions. Designers may make different choices to combine the peripheral functions with the platform ones to yield an SRR product family, based on their expertise or customer needs.

Particularly, the first nine platform functions listed in Table 2 constitute most of the functions of a popular SRR product in the market, the Sphero robot toy (Fig. 2(*a*)) from Sphero, Inc. The Sphero robot toy also includes some other functions, such as Emit Light and Wireless Charge. The company's later released BB-8 model (Fig. 2(*b*)) included more functions, such as Voice Control, Follow Path, Produce Sound (for sound effect), Hold and Pivot External Accessory (a rotating head) to mimic a robot in the Star Wars movie, and the BB-9E model (Fig. 2(*c*)) further included an augmented reality function to BB-8. Meanwhile, the first 13 core functions listed in Table 2 appear in Virgo (Fig. 2(*d*)), a platform robot developed in SUTD, on top of which more functions will be integrated for defense or entertainment. In addition to the 13 functions, Virgo also has functions such as Locate Target, Follow Path, Detect Obstacle, and Avoid Obstacle.

In practice, designers should take the core functions identified from the data-driven algorithm as a reference point, rather than a

[7]We carried out a sensitivity analysis on the obtained core-periphery coefficient by removing a certain function from or adding a certain function to the obtained core. Results show that removing from the core a function with a relatively high coreness (e.g., coreness ranking higher than 12 in Table 1) or adding to the core a function with a relatively low coreness (e.g., coreness ranking lower than 41 in Table 1) may decrease the core-periphery coefficient significantly (e.g., >5%). Meanwhile, removing a function from or adding a function to the core with a coreness ranking between 12 and 41 only lowers the coefficient slightly (<5%).

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Table 2 The list of core functions for an SRR platform design

Core functions	Description	Component choice
Roll on ground	Roll on ground in a self-propelled manner	Drive system, wheels
Covert energy	Convert stored energy to kinetic energy for driving	Electric motors
Store energy	Store energy	Battery
Control direction	Control the driving direction of the robot	Propelled wheels
Remote control	Remotely controlled by an external controller	Bluetooth
Transmit data	Send data to controller or display devices	Bluetooth
Process data	Process data for controlling the movement of the robot and fulfilling other functions	Microcontroller board
Store data	Store data, such as images and sounds	Memory
Self-balance	Keep Itself in a ready position for driving. For example, avoid the inner cart turning upside down	Supporting pod
Plan path	Plan a path to get a given target	Microcontroller board
Photograph surrounding	Generate images of the surroundings of the robot	Camera
Detect acceleration	Detect the acceleration of an inner part of the robot	Accelerometer (inertial measurement unit)
Detect angular velocity	Detect the angular velocity of an inner part of the robot	Gyroscope (inertial measurement unit)
Measure environment	Measure the surrounding environment of the robot, such as temperature, light intensity, radioactivity, and nuclear)	Sensors for measuring temperature, etc.
Track positions	Track the absolute positions of the robot	Global positioning system

rigid result. In the case of the SRRs, the designers or companies may include additional functions than the 15 core ones or a subset of them in a platform, by considering user and market needs, their own technical expertise, as well as the company's strategy. The designers may also examine the coreness scores or other measures of individual functions to decide whether to include it in the platform. The coreness score of each function resulting from the continuous model with its rank by the score in parentheses is also listed in Table 1. The core functions identified by the discrete model have the highest coreness scores. In Table 1, we italicize the functions appearing in Virgo and Sphero's product platform but not included in the core. Most of them have relatively high coreness scores as well. The coreness scores may provide another reference to weigh the functions for inclusion in a product platform for designers.

5 Discussion

In this paper, we present a data-driven method to identify the functions to be considered for inclusion into a product platform. The method applies core–periphery network analysis to a function network constructed based on the relationships of function co-occurrences in prior designs in a product or systems domain. In the case study of SRRs, 15 potential platform functions are identified using the network-based data-driven method. As suggested in Table 1, these core functions are among the most frequently occurring ones in prior designs, but not always. For example, the function, Propel in Water, has a high occurrence (ranking at the eighth) but is founded not in the network core. In particular, our method provides an algorithm-based stopping point in the search for functions to include in a platform. The resultant dichotomous partition of core and peripheral functions may serve as a reference point, based on which designers may add some peripheral functions or remove some core functions for an eventual platform design according to their expertise, customer needs, and specific strategies. Apart from the discrete core–periphery partition, the coreness scores from a continuous model also provide additional reference for designers to make an adjustment and a final decision.

In our case study, we include all the related patents in the analysis and treated all of them the same. In practice, designers may classify the prior designs and then identify the platform functions for different market niches. For example, designers may analyze only the SRR patents classified in the international patent class A63 "Sports; Games; Amusements"[8] to develop a platform for toys or analyze the SRR patents classified in B62 "Land Vehicles for Traveling Otherwise than on Rails" to design a platform for a family of SRR products functioning as vehicles. Meanwhile, designers can consider and select different subsets of peripheral functions integrated with the shared set of platform functions to generate variants in a product family. Designers may further categorize the peripheral functions into various groups satisfying different market niches with information about the application fields of prior designs, so that the differentiating functions can be selected more conveniently and purposefully.

Meanwhile, functions are the intermediary through which user or customer needs and product-system features are translated into design parameters in practice [107]. Functions identified from products have been used for long to abstract design tasks and model product architectures and product family architectures [16,90,108]. The data-driven method for identifying platform functions can serve as a prior step before identifying modules from a product architecture through function-based approaches, such as MFD approach [14,109], functional-model approach [16,110], function-component-matrix approach [19], and dendrogram approach [18]. Specifically, the identified platform functions can be further decomposed into secondary functions to inform the design of modules within the platform. Note that, given a specific platform design according to primary functions, designers may still make varied choices for secondary functions, components, design parameters, etc.

In brief, the proposed data-driven method differentiates itself from others by using an empirical function co-occurrence network to represent the design knowledge extracted from prior designs and detecting the core–periphery structure in function networks to identify the core functions for platforming considerations. The method is based on massive prior design data instead of randomly collected user requirement data or a confined set of existing variants. We show in the case study that the massively and readily available patent data are suitable for and can empower this network analysis method. Meanwhile, it is focused on functions, and thus, complements other platform analysis methods focused on components, modules, and design parameters. In addition, with continual advances in artificial intelligence, the presented method can be automated easily and serve as a computer-based aid to complement other heuristic and small-sample methods for platform planning.

It is noteworthy that several innovative design methods, such as TRIZ [111] and design-by-analogy [112], have previously been proposed to utilize design knowledge contained in patents. In general, these methods focus on developing solutions to specific design problems, but do not address product platform identification. This research presents a first effort to exploit design

[8] http://www.wipo.int/ipc/itos4ipc/ITSupport_and_download_area/20100101/pdf/scheme/core/en/ipcr_en_a_core_2010.pdf

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knowledge in patents for product platform design. Also, several studies have applied optimization algorithms to product platform and family analyses, for example, for finding the optimal commonality of a product family from cost and performance perspectives [20,32,33]. Our method stands out by focusing on functions co-occurrences in prior designs and finding optimal partition of core and peripheral functions in the function co-occurrence network to inform platform design.

The present paper has a number of limitations that may be addressed in future work. The first is that rich data on prior designs in the product domains are required. In this study, we only demonstrate the use of patent data. Other forms of data on prior designs in a field should be explored as well. Second, highly novel or radically innovative products would not have many precedent designs. The method is most useful for platform identification when a field or a firm has accumulated many designs, when platform analogies are sought for known domains, or for the re-redesigns of an existing platform as design data continue to grow and the design knowledge base expands. In addition, we only present the case study of SRRs in this paper. In future research, we anticipate applying this method to other platform design cases, to better understand the contextual factors that influence the choice of core–periphery models and the effectiveness of the method.

6 Conclusions

In this paper, we present a data-driven method to identify the functions for inclusion in a potential product platform. The method is centered on detecting the optimal core–periphery partition in the function co-occurrence network, which is constructed based on the function data in prior designs of a product or systems domain. We demonstrate the method by applying it to identify functions in the network core to inform the platform design of spherical rolling robots based on the relevant patent data and to form product family by integrating the platform functions with varied sets of peripheral functions. In brief, the data-driven method assists in product platform and family design and complements the existing heuristic and qualitative approaches of platform planning.

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